THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A lake in which the algae grow excessively after fertiliser run-off will show: 1

(a) a decrease in dissolved oxygen after the algae die
(b) cool water
(c) clear water
(d) increase in fishes

2. In which of the following would the loss of vultures have the greatest effect? 1

(a) nitrogen fixation **(b)** pollination
(c) seed dispersal **(d)** removal of dead animals

3. Assertion (A): Plants are called producers. Reason (R): They obtain food by eating other organisms. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The population of an herbivore falls in a grassland. Which change is LEAST likely? 1

(a) decline in energy at the next level **(b)** increase of the predators
(c) increase of grass **(d)** decrease of the predators

5. Predators help prey populations by: 1

(a) causing extinction always **(b)** feeding them
(c) protecting them **(d)** keeping their numbers in check

6. Assertion (A): A food web is more stable than a food chain. Reason (R): A food web has alternate pathways for energy flow. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Which of the following correctly shows the direction of energy flow? 1

(a) grass to grasshopper to frog **(b)** frog to grasshopper to grass
(c) grasshopper to grass to Sun **(d)** Sun to frog to grass

8. An example of competition in an ecosystem is: 1

(a) bees pollinating **(b)** plants making food
(c) bacteria decomposing **(d)** two animals fighting for the same food

9. Assertion (A): The number of organisms decreases as we go from producers to top carnivores. Reason (R): Energy available decreases at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A student observes that the sum of the biomass of herbivores is much less than that of plants in a forest. This supports: 1

(a) decomposers eat all
(b) loss of energy between levels
(c) no herbivores
(d) producers are small

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A farmer grows pulses and cereals alternately. Explain the benefit. 2

12. How does afforestation help in controlling floods? 2

13. Why are the grassland and forest food chains in the tropics longer than in the polar regions? 2

14. What could be the effect of killing all the snakes in a field? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the loss of habitat leads to the extinction of species with the example of the tiger. 3

16. Explain why the pollution caused by plastics persists for hundreds of years. 3

17. A forest fire destroys vegetation. Describe the stages of recovery of the ecosystem. 3

OR

Explain the role of top predators in controlling herbivore populations and their effect on vegetation. 3

18. Explain how global warming may affect polar bears, coral reefs and farm crops. 3

OR

Explain the role of urban green spaces in air quality and human health. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the cycling of matter and the flow of energy in an ecosystem, bringing out the difference between them with examples. 5

OR

Describe the water cycle and explain how deforestation, urbanisation and rainwater harvesting each change it in a locality. 5

20. Discuss any five major threats to biodiversity in India and the conservation strategies adopted. 5

OR

Describe the interdependence between humans and nature with five examples, and the consequences when the relationship is broken. 5

END OF QUESTION PAPER — Total: 40 Marks